APEKSHA MORAPPANAVAR

Software Engineer | 1 Year of Experience | C++ · Python · Backend · Distributed Systems

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Software Engineer with 1+ year of production software development experience in C++ and Python, with hands-on backend/full-stack development using Node.js, Express, MongoDB, REST APIs, and TypeScript. Experienced in software design, debugging, automated testing, performance optimization, and building multi-tenant backend systems. Strong foundation in DSA, OOP, Linux, multithreading, and distributed systems.

WORK EXPERIENCE

Software Engineer | *Magna Automotive India Pvt Ltd, Bangalore* March 2025 - Present

- Developed and modernized object-oriented C++ software components from system requirements, migrating legacy implementations to maintainable architectures while improving code quality and readability.
- Designed and executed 50+ unit tests using Google Test (gTest), strengthening software correctness and supporting regression testing practices.
- Built 3+ Python automation tools for data processing, result visualization, and log conversion, reducing manual effort in validation workflows.
- Performed performance profiling and optimization, achieving ~40% improvement in runtime efficiency over baseline implementations.
- Collaborated with cross-functional engineering teams to deliver production-ready software while maintaining design, testing, and documentation standards.
- Performed static code analysis using QA-C++ to identify and resolve coding-standard violations and maintainability issues.

PROJECTS

MultiFlix — Full-Stack Multi-Tenant Media Platform | *Personal Project | GitHub*

- Built a full-stack media platform with a multi-tenant architecture where each billing account manages multiple users, roles, and isolated profiles.
- Implemented JWT/Passport.js authentication and role-based access control (RBAC), distinguishing owner and member permissions across tenant accounts.
- Developed profile-scoped personalization including watch history, watchlist, continue-watching, and recommendation generation using TMDb data.
- Designed and integrated REST APIs for content discovery, Google OAuth login, and invite-based onboarding, with MongoDB/Mongoose-backed persistence.

MERN-GPT — AI-Powered Full-Stack Chat Application | *Personal Project | GitHub*

- Built a full-stack conversational application using React, Node.js, Express, and MongoDB, integrating the Gemini API to generate AI-powered responses.
- Implemented persistent chat storage and a React interface using Vite, Material UI, and Context API for conversational state management.

Breast Cancer Detection Using Few-Shot Learning | *Research Project*

- Built a Prototypical Networks model with a 4-layer CNN encoder and data augmentation on the BUSI dataset for three-class ultrasound image classification.
- Achieved 85.75%–91.67% accuracy across 5/10/15-shot settings, outperforming VGG-16 and Inception V3.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science and Engineering | *KLE Technological University* 2021 - 2025

CGPA: 9.44

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL

Backend & APIs: Node.js, Express.js, REST API Design, MongoDB, Mongoose, JWT & Passport.js Authentication, OAuth (Google)

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming (OOP), Distributed Systems, System Design, Multithreading, Linux, Git, Debugging, Optimization & Complexity Analysis, Unit & Integration Testing (Google Test) **Web / Full-Stack:** React.js, Vite, Tailwind CSS, HTML, CSS, Role-Based Access Control (RBAC), Web Development, Software Design & Architecture

AI / ML: Gemini API, CNNs, Few-Shot Learning, Data Augmentation, Model Evaluation

Other Tools: MATLAB, ROS2, Helix QAC (static analysis), Oracle SQL, Informatica Power Center, Windows

ACHIEVEMENTS

- Participated in the Smart India Hackathon and won the internal hackathon round at the institute level.
- Presented research on prototypical networks and few-shot learning for disease detection at INCET 2024.

CERTIFICATIONS

- Linear Programming and Its Applications to Computer Science — IIT Kanpur
- Breast Cancer Detection by Prototypical Networks Using Few-Shot Learning — IEEE Bangalore
- JNCIA-Junos (Juniper Networks Certified Associate – Junos) — Juniper Networks